Venkata Sai

Senior Cloud Infrastructure Engineer

venkatasai.jobs@gmail.com | +1(475)-291-1455 | Ann Arbor, MI

Professional Experience

Software Engineer *Jul 2025 – Present*

Epik Solutions | Enterprise Platform

- Designed and implemented multi-account **AWS Cloud** infrastructure using **Terraform** modules for a secure environment, streamlining resource provisioning with **CI-driven plan/apply workflows** by 40%.
- Managed and optimized **Kubernetes** clusters on **EKS** in **production environments**, configuring **cloud networking** including **VPCs** and **load balancing**, improving application uptime by 15%.
- Developed and enforced **least-privilege IAM** policies across **AWS** services, integrating **security groups** and **compliance tooling** to harden infrastructure and pass audits with 100% adherence.
- Created extensive **Python** scripts for operational **automation** and incident response, deploying serverless functions via **AWS Lambda** and managing data in **S3**, reducing manual toil by 25%.
- Implemented advanced monitoring and logging solutions for **EKS** clusters using **Prometheus** and **Grafana**, enabling proactive troubleshooting and reducing critical incident resolution time by 30%.
- Configured complex **VPC peering** connections and managed **DNS** for cross-region disaster recovery, ensuring resilient **cloud networking** that maintained 99.99% availability.
- Drove the adoption of **Infrastructure as Code** principles using **Terraform** and **Git** for all infrastructure changes, enhancing version control and **CI/CD** pipeline reliability by 20%.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | B2B Platform

- Developed and maintained serverless microservices using **AWS Lambda** and **Python**, enabling critical business logic and process **automation** for a large-scale enterprise **SaaS** platform, boosting efficiency by 18%.
- Deployed and managed applications on **Kubernetes** clusters provisioned on **AWS EKS** and **EC2** instances, ensuring high availability and scalability for customer-facing services in **production environments**.
- Wrote and refined **Terraform** configurations for provisioning **AWS** resources, including **S3** buckets and databases, integrating into **CI/CD** pipelines for consistent infrastructure deployments, speeding delivery by 15%.
- Implemented robust access controls using **IAM** roles and **security groups** within isolated **VPC** networks, adhering to **least-privilege** principles and fortifying application security posture by 20%.
- Created custom **Python scripting** for routine operational tasks, automating log analysis with **CloudWatch** and reducing manual data processing time by 30% for support teams.
- Configured and managed internal **cloud networking** components like **load balancing** with ALBs and **DNS** records, ensuring efficient traffic routing within a complex **VPC** setup, improving latency by 10%.
- Containerized applications using **Docker** and orchestrated deployments on **Kubernetes**, integrating into **CI/CD** pipelines to accelerate feature releases and maintain stability in **production environments** by 25%.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | SaaS Platform

- Developed backend services in **Python**, implementing **REST** APIs and creating efficient **scripting** for data processing tasks, improving system performance and facilitating client solutions by 20%.
- Engineered and deployed **Java Spring Boot** microservices for various client projects, handling data persistence with **PostgreSQL** databases and improving system responsiveness by 15%.
- Managed source code with **Git** and configured **CI/CD** pipelines using **Jenkins** for automated build and deployment processes, reducing manual errors and accelerating software delivery by 10%.
- Deployed and maintained client **web applications** on **AWS EC2** instances, utilizing **S3** for static assets and ensuring reliable service delivery with 99.9% uptime.
- Implemented basic monitoring and logging for applications, performing systematic **log analysis** to identify issues, improving system stability and reducing bug resolution time by 15%.

Skills

Languages: Python, Java
Frameworks & Tools: Terraform, Kubernetes, Docker, Jenkins, Git, Grafana, Prometheus, Spring Boot
Cloud & Infra: AWS Cloud, IAM, VPC, EC2, S3, EKS, Lambda, CloudWatch, multi-account environment, cloud networking, security groups, peering, DNS
Databases & Search: PostgreSQL
Concepts: Infrastructure as Code, CI/CD, Microservices, REST, Security Fundamentals, Networking, Least Privilege IAM, Compliance Tooling, Production Environments, CI-driven Plan/Apply Workflows, Automation, Scripting, Load Balancing

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India